CHAPTER

3 Bayesian Networks

Bayesian networks describe a high-dimensional probability model (a model over many random variables) in terms of a directed graph which encapsulates an expansion of the joint probability into a product of conditional probabilities.

3.1 Basic semantics

Let us suppose that we wish to specify a probability model $p(X)$ over a large set of random variables $X = (X_1, X_2, \ldots, X_n)$. For the moment, we will assume that the X_i are discrete, taking on finitely many values in some set $\mathcal{X}_i$. In general, specifying $p(X)$ requires a table over the configurations of X; if each $|\mathcal{X}_i| = d$, the table is of size d^n, and has $d^n - 1$ free parameters (since the probability distribution must sum to one).

Any probability distribution over $X_1, \ldots, X_n$ can be represented in terms of a series of conditional distributions, through application of the chain rule of probability. For example:

$$p(X_1, \ldots, X_n) = p(X_1)\, p(X_2|X_1)\, p(X_3|X_1, X_2) \ldots p(X_n|X_1, \ldots, X_{n-1})$$

Note that this expansion does not change the complexity of the representation; $p(X_1)$ specifies $d-1$ free parameters; $p(X_2|X_1)$ specifies $d(d-1)$ (since each of the d values of X_1 specify a distribution over X_2 with $d-1$ free parameters), and so on, giving

$$d - 1 + d(d-1) + d^2(d-1) + \ldots + d^{n-1}(d-1) = \frac{d^n - 1}{d-1}(d-1) = d^n - 1$$

free parameters.

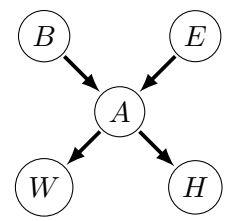
However, conditional independence can arise if some of these probability distributions do not depend directly on all the variables in their conditioning sets; for example, if $p(X_3|X_1, X_2) = p(X_3|X_2)$ (so that X_3 is independent of X_1 given X_2), then specifying the conditional distribution over X_3 will require fewer parameters.

We can draw a graph $G = (V, E)$ that represents the conditional independence structure of our expansion. To each variable X_i we associate a node $i \in V$ in the graph, and draw a directed edge from X_i to X_j, so that $(i \to j) \in E$, if X_j's conditional distribution depends on X_i. The resulting graph will be a *directed acyclic graph* (or DAG), since each variable can only depend on variables that precede it in

the conditioning order. We define the *parents* of X_j to be those nodes i such that edge $(i \to j) \in E$, and denote this set $\mathrm{pa}(j)$.

Example 3-1 : Burglar Alarm (Pearl, 1988)

Mr. Holmes has a burglar alarm in his house, but unfortunately his alarm is also prone to false alarms due to earthquakes. When he is at work, his two neighbors, Dr. Watson and Mrs. Hudson, will usually call him to tell him if the alarm is going off. However, they may also call by mistake, or even as a practical joke.



Let us define a Bernoulli (binary-valued) random variable $B \in \{0, 1\}$ to be the event that a burglary takes place, and similarly $E \in \{0, 1\}$ an earthquake, $A \in \{0, 1\}$ that the alarm goes off, $W \in \{0, 1\}$ that Dr. Watson calls, and $H \in \{0, 1\}$ that Mrs. Hudson calls.

Reasoning about the conditional distributions of a model for this problem, we might conclude that earthquakes and burglaries are independent of each other; the alarm going off is caused by either an earthquake or a burglary; and that Watson and Hudson's phone calls depend on whether the alarm is going off, but not on each other. These assumptions allow us to simplify the conditional expansion to:

$$p(B, E, A, W, H) = p(B)\, p(E)\, p(A|B, E)\, p(W|A)\, p(H|A)$$

Here we see that we have simplified $p(E|B) = p(E)$, due to the assumed independence of earthquakes and burglaries; and $p(W|A, B, E) = p(W|A)$ and $p(H|A, B, E, W) = p(H|A)$, since our two callers depend only on whether the alarm is sounding.

To fully specify the model, we must also specify the values of the required conditional probabilities, for example:

$p(B)$	$B = 0$	$B = 1$
	0.999	0.001

$p(E)$	$E = 0$	$E = 1$
	0.998	0.002

$p(A\|B, E)$	$A = 0$	$A = 1$
$BE = 00$	0.999	0.001
$BE = 01$	0.25	0.75
$BE = 10$	0.05	0.95
$BE = 11$	0.02	0.98

$p(W\|A)$	$W = 0$	$W = 1$
$A = 0$	0.95	0.05
$A = 1$	0.1	0.9

$p(H\|A)$	$H = 0$	$H = 1$
$A = 0$	0.99	0.01
$A = 1$	0.3	0.7

Representing the joint distribution

Under the assumptions that underly our model, we are able to represent and evaluate the joint probabilty distribution over the configurations of (B, E, A, W, H) much more efficiently. For example, although the joint probability specifies a probability for all 2^5 configurations, our model assumptions allow us to represent that joint probability using only $1 + 1 + 4 + 2 + 2 = 10$ values.

Example 3-2 : Joint distributions

The joint probability $p(B = 0, E = 1, A = 1, W = 1, H = 0)$ is given by

$$\begin{aligned} p(BEAWH = 01110) \\ &= p(B = 0)\, p(E = 1)\, p(A = 1|B = 0, E = 1)\, p(W = 1|A = 1)\, p(H = 0|A = 1) \\ &= 0.999 \cdot 0.002 \cdot 0.75 \cdot 0.9 \cdot 0.3 \end{aligned}$$

We can fill in all 2^5 entries of the joint probability, if desired:

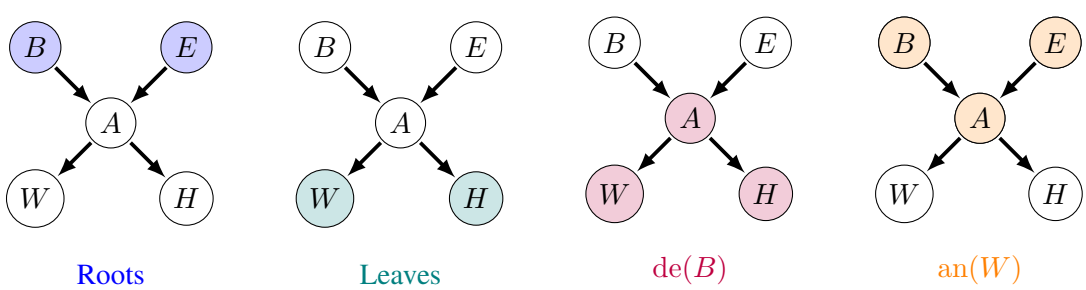
$BEAWH$	$p(B, E, A, W, H)$
00000	0.9367
00001	0.0095
00010	0.0493
00011	0.0005
$\vdots$	$\vdots$
11111	1.25×10^{-6}

Graph notation

It is useful to define some simple terms to discuss the structure of a Bayesian network. We have already defined the *parents* of X_i, denoted $\mathrm{pa}(i)$, to be the set of variables on which X_i's conditional distribution depends; in our graph, this is exactly the set of nodes j such that there is an edge from j to i: $(j \to i) \in E$. Conversely, the nodes $\{c : (i \to c) \in E\}$ are called the *children* of i. If X_i has no parents, we say that node i is a *root* of the Bayesian network; similarly, if it has no children, we say that it is a *leaf*. We define the *ancestors* of X_i, $\mathrm{an}(i)$, to be the set of all nodes a that have a directed path from a to i: so, i's parents, parents of parents, etc. Similarly, the *descendants* of i are those nodes d with a directed path from i to d.

Example 3-3 : Ancestors and descendants

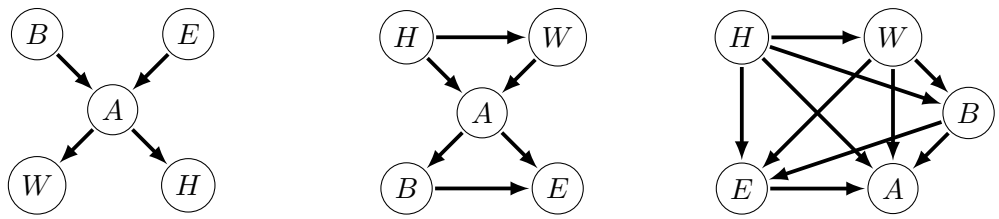
In the burglary network, B and E are roots, while W and H are leaves. The descendants of B are $\mathrm{de}(B) = \{A, W, H\}$, while the ancestors of W are $\mathrm{an}(W) = \{B, E, A\}$:



It is also sometimes useful to reason about *undirected* paths in the graph G, i.e., sequences of nodes that are connected by edges without regard to the direction of the edge. For example, the sub-graph $B \to A \to H$ is both a directed and an undirected path, while the sub-graph $B \to A \leftarrow E$ forms an undirected path, but not a directed one.

Conditional expansion order

The order in which we apply the rule of conditioning plays a significant role in the compactness of our distribution's representation. While any order of conditioning is equally valid, some will result in more or less simplification of the resulting conditional probability distributions.



(a) Order B, E, A, W, H (b) Order H, W, A, B, E (c) Order H, W, A, B, E

Figure 3.1: Using different orderings for the conditional expansion results in different Bayesian networks, depending on how the conditional probabilities simplify. In general, "causal" orderings (burglaries or earthquakes cause the alarm to sound, etc.) as in (a) are easier and simpler than non-causal orders like (b) (Mrs. Hudson and Watson do not cause the alarm; they merely provide information about it). Some orders may even result in conditional probabilities that do not simplify, such as the order in (c).

For example, let us assume that the joint distribution we constructed earlier correctly characterizes the burglary problem. However, we could just as easily write the alternative conditional expansion,

$$p(B, E, A, W, H) = p(H)\, p(W|H)\, p(A|H, W)\, p(B|H, W, A)\, p(E|H, W, A, B)$$

and then attempt to simplify each of these conditional probabilities. However, computing $p(W|H)$, we would find that the variables W and H are not independent – knowing that Mrs. Hudson has called means that it is more likely there is an alarm, and thus that Dr. Watson will call. **Add numerical values** Similarly, examining $p(A|W, H)$, we find that A is not conditionally independent of W or H given the other – for example, if both neighbors call, it provides more evidence that the alarm is truly going off than if either one called alone. We can, however, simplify $p(B|H, W, A) = p(B|A)$, since neither Dr. Watson nor Mrs. Hudson have any more information about a possible burglary beyond the alarm sounding. And finally, we find that $p(E|H, W, A, B) = p(E|A, B)$ for the same reason, but that E is not conditionally independent of either A or B given the other. So, putting this together, we have

$$p(B, E, A, W, H) = p(H)\, p(W|H)\, p(A|H, W)\, p(B|A)\, p(E|A, B).$$

In general, the conditional semantics of Bayesian networks lend themselves well to reasoning about "causal" interactions – burglaries do not cause earthquakes, but both can cause the alarm to go off, and so on. "Non-causal" orderings, such as H, W, A, B, E, make it significantly harder to identify independencies, and generally result in larger representations (here, $1 + 2 + 4 + 2 + 4 = 13$ parameters, compared to 10 for the original ordering). Some orders may even fail to result in any independencies; for example, if we expand the conditional probability along the order H, W, B, E, A, we would find that none of the conditional probabilities can be simplified – then, the resulting Bayesian network is the complete graph, in which each node connects to all its predecessors. This effect is illustrated in Figure 3.1.

Observations and influence

Using the earthquake example, we can see that reasoning about how probabilities are changed can be visualized using "paths" on the graph. For example, by brute force calculation, we can compute the marginal probability of the alarm going off, $p(A = 1) \approx 0.0025$. Some probabilities are more readily accessible from the tables; for example, it is easy to see that the marginal probability of an earthquake is $p(E = 1) \approx 0.002$

Now suppose that Dr. Watson calls. Observing his call ($W = 1$) makes it more likely that the alarm is going off, $P(A = 1|W = 1) \approx 0.043$, which in turn means that an earthquake is now more probable

than before our observation, $P(E = 1|W = 1) \approx 0.011$. The graph represents direct influence of one random variable on another: if earthquakes cause the alarm to go off, which causes Watson to call, then observing Watson's call (the effect) influences our belief about the probabilities of those causes.

3.2 Independence relations

One important use of the Bayesian network graph is the ability to reason about when independence or conditional independence holds for the distribution. In using only the graph, we are reasoning about independence relations that *must* hold, irrespective of the actual values in the conditional probability tables; equivalently, we can ask whether there exist *any* values for these tables for which the independence relationship will not hold. We saw that we can interpret the influence of observations as propogating along the graph; in this section, we will make this concept precise in order to reason about independence.

Let G be the graph of a Bayesian network. We define an (undirected) path $i_1 - i_2 - \ldots - i_L$ in G to be *active* given a set of "evidence" or observed variables $\mathcal{E}$ if and only if, for each node i_l:

1. The edges in G are $i_{l-1} \rightarrow i_l \leftarrow i_{l+1}$, i.e., both $X_{i_{l-1}}$ and $X_{i_{l+1}}$ are parents of X_{i_l}, and X_{i_l} or one of its descendants is in $\mathcal{E}$, or

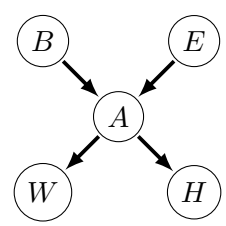
2. At least one of $X_{i_{l-1}}$ or $X_{i_{l+1}}$ is not a parent of X_{i_l}, and X_{i_l} is not in $\mathcal{E}$.

Then, given three sets of nodes, $\mathcal{A}$, $\mathcal{B}$, $\mathcal{E}$, we say that $\mathcal{A}$ and $\mathcal{B}$ are **d-separated** given $\mathcal{E}$ if there are no active (undirected) paths between any $X_i \in \mathcal{A}$ and $X_j \in \mathcal{B}$ given $\mathcal{E}$.

The concept of d-separation is fundamental to independence in Bayesian networks: if $\mathcal{A}$ and $\mathcal{B}$ are d-separated in G given $\mathcal{E}$, then $X_{\mathcal{A}}$ and $X_{\mathcal{B}}$ are conditionally independent given $X_{\mathcal{E}}$ in any $p(X)$ that can be factored according to G. Conversely, if they are not d-separated, then there exists some $p(X)$ with the factorization in G, and some value $X_{\mathcal{E}} = x_{\mathcal{E}}$, for which they are not conditionally independent.

Example 3-4 : Burglar Alarm and D-Separation

We can see the concepts of d-separation reflected in our example burglary Bayesian network.



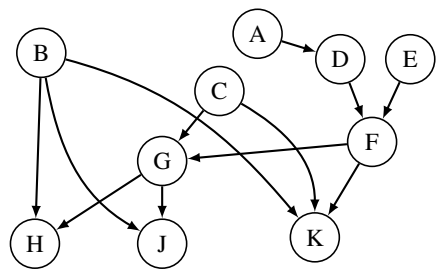
Suppose first that no information has been observed, so $\mathcal{E} = \varnothing$. Then, we can see that (as expected), burglary B and earthquake E are independent – the path between them passes through a "vee" shape ($B \rightarrow A \leftarrow E$), but neither A nor its descendants are observed ($\mathcal{E}$ is empty), so the path is not active. However, given an observation (say, that the alarm is on, or indirect evidence of this through one of the phone calls), B and E are no longer independent, since it is now more likely that at least one of them is true.

On the other hand, d-separation tells us that E and W are not independent (observing $W = 1$ makes E more likely). But, if we observe the value of the alarm, A, then the path $E \rightarrow A \rightarrow W$ is interrupted and becomes inactive, indicating that W has no more information beyond A's about E. Similarly, W and H are not independent (the path connecting them is active), since they are likely to co-occur as both are caused by the alarm going off. However, observing A makes the path inactive, and they are seen to be conditionally independent.

When there is more than one path between two sets of nodes, we can enumerate over the paths and check their status (active or inactive) to verify d-separation.

Example 3-5 : More D-Separation

Consider the Bayes net at right. There are several paths that connect B and E; for example:



$$B \to K \leftarrow F \leftarrow E$$
$$B \to J \leftarrow G \leftarrow F \leftarrow E$$
$$B \to H \leftarrow G \leftarrow F \leftarrow E$$
$$\vdots$$

Using d-separation, we can quickly see that:

- B is (marginally) independent of E (all paths involve $B \to X \leftarrow Y$)
- B is not independent of E given K ($B \to K \leftarrow F \leftarrow E$ active)
- B is not independent of E given H ($B \to H \leftarrow G \leftarrow F \leftarrow E$ active)
- B is independent of E given F (all paths $B \ldots \leftarrow F \leftarrow E$ inactive)
- B is independent of E given H, K, and F (same argument as just given F)

Equivalent graphs

D-separation tells us what independence relationships must hold for a given Bayesian network. However, as we saw in Figure 3.1, a given probability distribution may be represented using many possible directed graphs, depending on the order of expansion; depending on the resulting graph, some independencies may no longer be visible through d-separation. For example, while many independence relationships are visible from the graph in Figure 3.1(a), the graph in Figure 3.1(b) shows fewer independencies through separation, while Figure 3.1(c) shows none.

We say two graphs are **equivalent** (or more precisely, independence-equivalent, I-equivalent, or Markov equivalent) if the two graphs encode exactly the same set of independence and conditional independence relationships.

Example 3-6 : Equivalent graphs

Consider the following Bayesian networks:

(a) (b) (c) (d)

While each graph represents a different conditional factorization, graphs (a)–(c) are equivalent: their d-separation property implies the conditional independence of A and C given B, but no other independence relationships. On the other hand, (d) is *not* equivalent to (a)–(c), since it implies that A and C are marginally independent (i.e., given no observations), but *not* independent given B.

The notion of equivalent graphs becomes important when *learning* a Bayesian network from data. In particular, given any graphical model G_1 and its conditional probability distributions $p(X_j|X_{\text{pa}(()j)})$, if the graph G_2 is equivalent to G_1, then there exist conditional probabilitys for G_2 that make G_2's joint probability distribution equal to that of G_1's. This means that we cannot distinguish between G_1 and G_2 solely on the basis of their ability to fit a collection of observed data – any distribution of the data that G_1 can represent is also a possibile explanation using G_2. To give a simple example, since $A \to B$ and $B \to A$ are equivalent, we cannot tell whether A causes B, or B causes A, simply from observations; the observations can only tell us that A and B are correlated (depend on each other in some way).[1]

3.3 Learning from complete data

In practice, we often want to fit, or learn, a model from some observed data $D = \{x^{(j)}\}$. Bayesian networks allow us to define probability distributions over high-dimensional sets of random variables in a compact way. For the time being, let us assume that the *structure* of the Bayesian network (i.e., what independence relationships we wish it to enforce) is known, but not the actual probabilities, and we would like to estimate these probabilities from the data D. Furthermore, we will assume that the dataset D is **complete**, i.e., that each data point $x^{(j)}$ contains a measurement value for each of the n variables, $X = [X_1, \ldots, X_n]$. (We postpone the more difficult problems of estimating the structure of a graphical model to Chapter **??**, and of working with missing data values to Chapter 9.)

Given the structure of the Bayesian network (the directed graph), we can parameterize the distribution in terms of the conditional probabilities $p(X_i|X_{\text{pa}(i)})$, e.g.,

$$p(X_i = x_i|X_{\text{pa}(i)} = x_{\text{pa}(i)}) = \rho_{i,x_i,x_{\text{pa}(i)}}$$

Then, we can find the maximum likelihood estimates of the parameters; the data likelihood is given by,

$$\begin{aligned}\mathcal{L} &= \sum_j \log p(x^{(j)}\,;\,\rho) = \sum_j \sum_i \log p(x_i^{(j)}|x_{\text{pa}(i)}^{(j)}\,;\,\rho) \\ &= \sum_i \sum_{x_{\text{pa}(i)}} \sum_{j:x_{\text{pa}(i)}^{(j)}=x_{\text{pa}(i)}} \log p(x_i^{(j)}|x_{\text{pa}(i)}\,;\,\rho)\end{aligned}$$

The first equality uses the structure of the network to factor $\log p(X;\rho)$ into a sum with one term per variable X_i. Then, the last equality reverses the sum over i and j, and re-organizes the sum over data points (index j) to group the data by the value of their parent set's configuration $x_{\text{pa}(i)}$. This grouping allows us to optimize our model parameters independently – specifically, each term,

$$\sum_{j:x_{\text{pa}(i)}^{(j)}=x_{\text{pa}(i)}} \log p(x_i^{(j)}|x_{\text{pa}(i)}\,;\,\rho)$$

is the log-likelihood of a discrete distribution over X_i, using only the data in which we have observed that our parent set $X_{\text{pa}(i)}$ took on value $x_{\text{pa}(i)}$. This likelihood involves only a single row of the conditional probability table $p(X_i|X_{\text{pa}(i)})$, and we know from Chapter 2 that its maximum likelihood estimate is

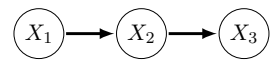
$$\hat{p}(X_i = x|X_{\text{pa}(i)} = x') = \frac{\#\{j : x_i^{(j)} = x, x_{\text{pa}(i)}^{(j)} = x'\}}{\#\{j : x_{\text{pa}(i)}^{(j)} = x'\}},$$

[1] To determine "causality", we require either additional assumptions about how two quantities A, B can influence one another (e.g., "A happened first, so it must have caused B"), or the ability to intercede and force one of A or B to take on a different value; this latter concept is made formal in causal reasoning such as `do`-calculus **?**.

i.e., the empirical fraction of the data with $X_i = x$, out of those data where $X_{\mathrm{pa}(i)} = x'$. As with the estimators discussed in Chapter 2, we can also apply a Bayesian prior or add regularization terms to the maximum likelihood estimator to reduce its potential for overfitting, for example by adding some small value $\alpha_{xx'}$ to the numerator, and $\sum_x \alpha_{xx'}$ to the denominator.

Example 3-7 : Fitting a Small Model

Suppose we wish to fit the Bayesian network shown at right using the small ($m = 10$) data set shown. We can compute the maximum likelihood estimates of the probabilities as,



Bayesian network

$\hat{p}_1(X_1)$	$X_1 = 0$	$X_1 = 1$
	2 / 10	8 / 10

$\hat{p}_{12}(X_2 \mid X_1)$	$X_2 = 0$	$X_2 = 1$
$X_1 = 0$	1 / 2	1 / 2
$X_1 = 1$	3 / 8	5 / 8

$\hat{p}_{23}(X_3 \mid X_2)$	$X_3 = 0$	$X_3 = 1$
$X_2 = 0$	2 / 4	2 / 4
$X_2 = 1$	2 / 6	4 / 6

X_1	X_2	X_3
1	0	0
1	1	1
1	1	1
1	0	1
1	1	1
1	1	0
0	1	0
1	1	1
1	0	0
0	0	1

Observed dataset

Compared to the empirical joint distribution, the conditional independence assumptions imposed by the graph's factorization reduce our tendancy to overfit to a small data set. For example, although the pattern $X = [0, 0, 0]$ is never explicitly observed in the data, our model estimates its probability as non-zero,

$$\begin{aligned} p(X = [000]) &= \hat{p}_1(X_1 = 0)\, \hat{p}_{12}(X_2 = 0 \mid X_1 = 0)\, \hat{p}_{23}(X_3 = 0 \mid X_2 = 0) \\ &= \quad (2/10) \quad \cdot \quad (1/2) \quad \cdot \quad (2/4) \\ &= 0.05, \end{aligned}$$

since we do observe some data corresponding to each of the conditional probability table entries.

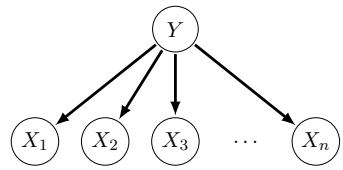
3.4 Example Bayesian networks

Several types of Bayesian network are sufficiently common special cases to deserve closer examination.

Naïve Bayes

Naïve Bayes models are a classic approach for classification (predicting a discrete label) from large numbers of observed features. Suppose, for example, that we wish to predict whether a driver will be in an accident during the next year. We might gather lots of information about the driver, which we can denote as observed variables X_i; for example, $X_1 = 1$ if the driver is a student; $X_2 = 1$ if the driver' age is over 25; and so forth. After observing the values of the X_i, we want to compute the probability $p(Y \mid x_1, \ldots, x_n)$ that this particular driver with attributes x will have an accident.

Unfortunately, for large n (for example, hundreds of features that might be relevant to driver risk), the joint distribution is too large to be practical, with $O(2^n)$ probabilities that need to be estimated and stored. Instead, we simplify this distribution by assuming a significant degree of conditional independence, namely that the target label Y (accident or not) is a "cause" that influences the observed features, but that the features are themselves conditionally independent of one another given Y. The joint distribution then factors as



Naïve Bayes model

$$p(Y, X_1, \ldots, X_n) = p(Y)\, p(X_1|Y)\, p(X_2|Y)\, \ldots p(X_n|Y),$$

leading to the Bayesian network representation shown at right.

Historically, this type of model was frequently used for spam filtering, by treating each word of the message as a feature X_i which is conditionally independent of the other words given the message type (spam or not). In fact, there are a number of formulations of naïve Bayes models for spam filtering, which vary in terms of the choice of the observed features X_i; see, for example, Metsis et al. (2006) for an overview of several. A typical version is to take X_i to be the ith word in the document, taking on values corresponding to all possible English words.

Example 3-8 : SMS Spam Filtering

Let us fit a simple naïve Bayes model to predict whether a message is spam or not, based on which words appear in its text. We use the *SMS Spam Collection* dataset (Almeida et al., 2011), which aggregates several sources of spam and non-spam ("ham") messages. From the labels, we estimate $\hat{p}(Y = \text{spam}) \approx 0.134$, i.e., about 13% of the dataset are spam messages. We pre-process the data to remove rare words and punctuation, and estimate the word probabilities in spam and non-spam; since some words may not appear in one category or the other, we regularize by estimating MAP probabilities and including a Beta prior, Beta$(\rho\,;\, a = b = 2)$ (i.e., add one pseudo-observation of each possible outcome). Some words ("the", "and") are common in both sets, while others are far more common in either spam ("prize") or non-spam messages ("later!"):

$$\hat{p}(\text{prize}|\text{spam}) = 0.00623 \qquad \hat{p}(\text{the}|\text{spam}) = 0.01386 \qquad \hat{p}(\text{later}|\text{spam}) = 0.00007$$
$$\hat{p}(\text{prize}|\text{ham}) = 0.00002 \qquad \hat{p}(\text{the}|\text{ham}) = 0.01788 \qquad \hat{p}(\text{later}|\text{ham}) = 0.00215$$

We can compute a "score" for each message corresponding to the log-odds of the two classes, e.g.,

$$\eta = \log \frac{p(\text{spam}|X)}{p(\text{ham}|X)} = \log \frac{p(\text{spam}, X)}{p(\text{ham}, X)} = \log \frac{p(\text{spam})}{p(\text{ham})} + \log \frac{p(X_1|\text{spam})}{p(X_1|\text{ham})} + \ldots + \log \frac{p(X_n|\text{spam})}{p(X_n|\text{ham})}$$

which depends on the observed words X_i. So, for example, given a message:

(prior)	WINNER!!	As	a	valued	network	customer	you	have	been	selected	to	receivea	£900	prize	reward!
-1.87	4.28	0.0	0.41	2.83	3.09	3.05	-0.43	0.22	0.74	3.09	0.63	0.0	0.0	5.98	3.84

where we mark large positive terms in red (significantly increase the likelihood the message is spam), large negative terms in green (significantly decrease the likelihood of spam), and rare/ignored words (such as the typo, "receivea") or approximately equally probable words in gray. In this case, we compute $\eta \approx 16$, and so $p(\text{spam}|X) = (1 + \exp(-16))^{-1} \approx 1$, almost certainly spam.

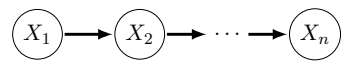
Taking another message, we have,

(prior)	Almost	there,	see	u	in	a	sec
-1.87	-1.19	-1.08	-0.51	-0.49	-0.96	0.41	0.0

totaling $\eta \approx -5.7$ and so $p(\text{spam}|X) = (1 + \exp(5.7))^{-1} \approx .003$, not likely to be spam.

Note that often, the features X_i are not truly even conditionally independent of one another; for example, there is considerable structure and correlation among words in emails or texts that is not being captured by the graph. However, if we are only interested in predicting Y, ignoring this structure may be an acceptable approximation. Next, we consider a type of Bayesian network that can be used to model sequential data, such as time series or natural language sentences.

Markov chains



Markov chain model

Perhaps the most common probability model for time series or other sequential data is the **Markov chain** model. We associate a random variable X_t (often called the **state variable**) with each time t, and assume that the state at time t depends only on the value of the previous state, X_{t-1}. Thus, the overall distribution of the sequence $(X_1, X_2, \ldots, X_n)$ can be written

$$p(X_1, X_2, \ldots, X_n) = p(X_1)\, p(X_2|X_1)\, p(X_3|X_2)\, \ldots\, p(X_n|X_{n-1})$$

In most Markov models, we assume that the distribution $p(X_t|X_{t-1})$ is the same for all times t, in which case the Markov model is called **homogeneous** and the distribution $p(X_t|X_{t-1})$ is called the **state transition distribution**.

It is easy to see that in the Markov chain model, any node X_t d-separates the nodes on its left from those on its right. This means that, given the state value $X_t = x_t$ at some time t, the future evolution of the system, $(X_{t+1}, \ldots, X_n)$, is independent of its past history $(X_1, \ldots, X_{t-1})$, i.e., x_t is a sufficient statistic for the entire past.

Markov models are a common tool for text data, which has a natural sequential structure. In a **bigram model**, the sequence probabilities depend on the pairs of adjacent words in the sentence. We can define the state X_t be the tth word, so that the transition distribution $p(X_t|X_{t-1})$ identifies which words commonly follow each other in a sequence.

Example 3-9 : Bigram Yelp Reviews

Using a corpus of five-star Yelp reviews (Yelp, 2021), we can build a model of the language of positive restaurant comments. For simplicity, we keep only the $d = 1000$ most common words, and estimate the transition probability by simply counting how often each word x_{t-1} is followed by any other word x_t. Normalizing these counts to sum to one, we find, for example,

$$\hat{p}(X_t|X_{t-1} = \text{great}) = \begin{cases} 0.054 & \text{place} \\ 0.053 & \text{food} \\ \vdots & \vdots \end{cases} \qquad \hat{p}(X_t|X_{t-1} = \text{ordered}) = \begin{cases} 0.413 & \text{the} \\ 0.131 & \text{a} \\ \vdots & \vdots \end{cases}$$

We also estimate the distribution of the first word (the *initial state*), $p(X_1)$. If desired, we can include a special "end of sequence" token, indicating the end of the review; this makes $p(X)$ a probability distribution over sequences of arbitrary length.

Once we have estimated the parameters of our Markov chain model, it is easy to draw samples from $p(X)$, by sampling $x_1 \sim p(X_1)$, then $p(X_2|X_1 = x_1)$, and so on, until we draw an end-of-sequence value. This leads to samples for X such as,

this guy and i recommend this place is awesome food wait for any idea how great reading the owners for me to remember to wow its own table . . .

In the resulting sample, we can see that each pair of words tends to make sense together, resulting in plausible short sequences that do not always make sense together. For example, the subsequences "i recommend this place," "this place is awesome," and "awesome food" are all likely, but when strung together by the limited, single-word Markov dependence may not make holistic sense.

N-gram models add longer-range temporal structure by making each X_t depend on the previous $n-1$ words in the sentence – so, a trigram model uses a distribution $p(X_t|X_{t-1}, X_{t-2})$, while a unigram model generates each word independently according to some $p(X_t)$.[2] Although the dependence becomes longer-range and more complex as n increases, conceptually n-gram models remain equivalent to a Markov chain for an appropriate state definition: let $\tilde{X}_t = [X_t, X_{t-1}]$ for each t; then the state transition

$$p(\tilde{X}_t = [x_t, x_{t-1}] \mid \tilde{X}_{t-1} = [x'_{t-1}, x'_{t-2}]) = p(X_t = x_t|X_{t-1} = x'_{t-1}, X_{t-2} = x'_{t-2})\, \delta(x_{t-1} = x'_{t-1})$$

defines a state transition from the preceding pair of words $\tilde{X}'_{t-1}$ to the current pair $\tilde{X}_t$, enforcing the constraint that the word on which the two overlap (position $t-1$) is identical in both. However, practically speaking this transformation may be inefficient, since if the number of possible values for each X_t is $|\mathcal{X}_t| = d$, a table representation of $p(\tilde{X}_t|\tilde{X}'_{t-1})$ would nominally have d^4 entries, although all but d^3 will be zero.

Given that there are many possible words x, we may not have enough data to accurately estimate $p(X_t|X_{t-1})$ (much less higher order n-gram models), particularly for rare words x_t. In such cases, it can be important to regularize or "smooth" the probability estimates, for example by adding regularization terms $\alpha_{xx'}$ to the empirical estimate of $p(X_t = x|X_{t-1} = x')$, corresponding to Dirichlet priors over the conditional probabilities. In practice, it can matter a great deal how these priors are selected. For example, selecting a prior in which all word transitions are equally likely, so that all the $\alpha_{xx'}$ equal the same constant, $\alpha_{xx'} \equiv \alpha_0$, is known as *Laplace smoothing* in natural language processing, but is often not very effective. A more sophisticated idea, called *Jelinek-Mercer smoothing* (Jelinek and Mercer, 1980), is to use estimated lower-order probabilities to define the prior, e.g., $\alpha_{xx'} = \alpha_0\hat{p}(x)$, where $\hat{p}(x)$ is a (possibly Bayesian) estimate of the unigram model of the data. This way, our prior implies that never-seen transitions from x' to common words x are more likely than transitions to rare words. *Kneser-Ney smoothing* (Ney et al., 1994) does something similar, but replaces the estimated probabilities $\hat{p}(x)$ with the number of unique words that are followed by x (normalized to sum to one). Then, even if x is common, but only appears after a few unique words, our prior will place low probability on new, unseen transitions to x. Conversely, if many distinct words all transition to x, additional, not-yet-seen transitions to x are more plausible.

Transformer models & LLMs; attention = compact dependence on long history of observed tokens.

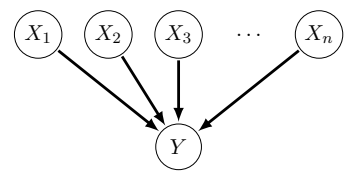
[2] Our naïve Bayes spam filter in Example 3-8 can be viewed as two unigram models, one for spam and one for non-spam.

Noisy-OR models

Bayesian networks provide an efficient representation of a probability distribution when there are significant conditional independencies in the model, i.e., when each "effect" (child) variable does not depend on too many "causes" (parents). Indeed, the standard representation of the conditional probability table, e.g., $p(X_c|X_{\mathrm{pa}(c)})$, has a number of rows that is exponential in the number of parents $|\mathrm{pa}(c)|$.

However, in some settings, we may have a variable that has a large number of possible causes. For example, our car failing to start could be caused by a low battery, low fuel, a bad starter, and so on. In this case, our child node Y has a large number of parents X_i, leading to an extremely large conditional probability table, $p(Y|X_1, \ldots, X_n)$[3].

One method of compactly defining a conditional probability with many parent variables is to assume it consists of a simple form with relatively few parameters. A classic example of this is the "noisy OR" model for Y, in which the variables $Y, \{X_i\}$ are all binary, $\{0, 1\}$, and we have a single probability ρ_i for each X_i with which we define,



Noisy OR model

$$p(Y = 1|X_1, \ldots, X_n) = 1 - \prod_i \rho_i^{\mathbb{1}[X_i=1]}$$

In other words, when all $X_i = 0$, we have $Y = 0$; if a single $X_i = 1$, then we see that $Y = 1$ with probability $1-\rho_i$; and as more X_i turn on, the chances that $Y = 1$ increase. If all the $\rho_i \to 0$, this reduces to a deterministic, boolean OR constraint. (A minor variation introduces an additional probability ρ_0 to the product, called a "leak", which allows the possibility that $Y = 1$ even when all underlying causes $X_i = 0$.)

Example 3-10 : Noisy OR

Suppose we have a variable H that indicates whether or not we have a headache, with three possible causes: C (whether we have a cold), F (whether we have the flu), and D (whether we are simply dehydrated). Let

$$\rho_C = 0.2 \qquad \rho_F = 0.1 \qquad \rho_D = 0.5$$

Then, the conditional probability table for H is given at right. Bold entries indicate where the probability value corresponds to one of the ρ_i, while products of the ρ_i are written out explicitly (e.g., $\rho_C \times \rho_F = (0.2) \times (0.1)$) to highlight their calculation, and $p(H = 1|C, F, D) = 1 - P(H = 0|C, F, D)$. As more potential causes are true, the probability of $H = 1$ increases. Although the full table specifies 8 (linearly independent) probabilities, the entries are defined using only three parameters.

$p(H\|C,F,D)$	$H=0$	$H=1$
$CFD = 000$	1.0	0.0
$CFD = 001$	**0.5**	0.5
$CFD = 010$	**0.1**	0.9
$CFD = 011$	$(0.5) \times (0.1)$	0.95
$CFD = 100$	**0.2**	0.8
$CFD = 101$	$(0.2) \times (0.5)$	0.9
$CFD = 110$	$(0.2) \times (0.1)$	0.98
$CFD = 111$	$(0.2) \times (0.1) \times (0.5)$	0.99

It is sometimes possible to represent systems with many possible causes efficiently by introducing extra variables that are used to simplify the structure of the graph. One example of such are latent variables (variables whose value cannot be directly observed), which we examine in more detail in Chapter 9. Others are simply convenient transformations of the representation. For example, the noisy OR model can be rewritten by introducing extra, auxilliary variables, in such a way that no variable has more than two parents, as we can illustrate:

[3]Notice, this is the converse of the naïve Bayes model, in which one parent had many children, leading to a compact representation.

Example 3-11 : Noisy OR (2)

Let us construct the same model as in Example 3-10. Define an auxilliary variable $\tilde{C}$ such that

$$p(\tilde{C} = 1 \,|\, C) = \begin{cases} 0 & C = 0 \\ 1 - \rho_C & C = 1 \end{cases},$$

and similarly define $\tilde{F}$ and $\tilde{D}$ using ρ_F and ρ_D.

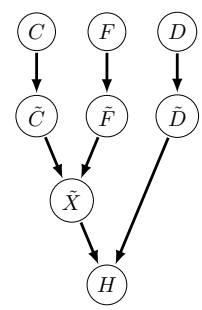
Next, define deterministic conditional distributions enforcing that $\tilde{X} = \tilde{C} \vee \tilde{F}$ and $H = \tilde{X} \vee \tilde{D}$ with probability one, i.e.,

$$p(\tilde{X} = 1 \,|\, \tilde{C}, \tilde{F}) = \begin{cases} 0 & \tilde{C} = \tilde{F} = 0 \\ 1 & \text{otherwise} \end{cases},$$

This gives a model (with graph illustrated at right) with no more than two parents per variable, and whose distribution

$$p(H|C, F, D) = \sum_{\tilde{c}, \tilde{f}, \tilde{d}, \tilde{x}} p(H, \tilde{c}, \tilde{f}, \tilde{d}, \tilde{x}|C, F, D)$$

is identical to that defined in Example 3-10.



Auxilliary variable noisy OR

Since the resulting model has at most two parents per variable, it can be represented using conditional probability tables of size 2^3, and the representation size of the model grows only linearly with the number of causes, $\{X_i\}$. However, transformations such as this may have implications for learning; if we can only observe the variables $[C, F, D, H]$ in our data set, the variables $[\tilde{C}, \tilde{F}, \tilde{D}]$ are "latent", or unobserved, and the resulting learning task is a type of *incomplete data* or *missing data* learning problem (see Chapter 9).

3.5 Reasoning about Causality

Fill in. Maybe move to latent variables?

Although cause and effect give a useful intuition about Bayesian networks, we've seen that the BN structure itself does not require any causal mechanisms; we were able to construct burglar alarm networks, for example, that did not follow a reasonable causal ordering. However, if we think that the model *does* provide a reasonable description of cause and effect, we can use it to reason about the effect of intervention in the model, i.e., forcing one or more variables to take on a desired value. Can reason about the effect of this operation: replace $p(X|paX)$ with intervention dist $\tilde{p}(X)$, altering the graph around the intervention X. Pearl's **do-calculus**. Often requires introducing additional *latent* or unobserved variables that may influence the distribution. Example. Resulting $p(Y|do(X))$ differs from $p(Y|X)$ due to latent, confounding influences on X.